

MSET002

Question 1.

[Tg:@abusat_760] $\sqrt{8+x} = x - 4$

What value of x satisfies the equation above?

- A) -3
- B) 1
- C) 8
- D) 17

Question 2.

[Tg:@abusat_760] Which of the following is equivalent to the expression $2(x+3)(x-5) + 4(x-2)(x-5)$?

- A) $6x^2 - 20x - 10$
- B) $6x^2 - 32x + 10$
- C) $6x^2 + 32x - 10$
- D) $6x^4 - 32x^2 - 39$

Question 3.

[Tg:@abusat_760] First-time customers at a dog-boarding facility must pay a onetime \$20 evaluation fee, as well as \$40 for each day and \$30 for each night the dog spends at the facility. Which of the following expressions best models the total cost, in dollars, for a first-time customer to board a dog at this facility for d days and n nights?

- A) $(40 + 30)(d + n) - 20$
- B) $(40 + 20)d + 320n$
- C) $(40 + 30)(d + n)$
- D) $10(4d + 3n + 2)$

Question 4.

[Tg:@abusat_760] If $a > 0$, which of the following is equivalent to $\frac{-8a^2}{-4a^4}$?

- A) $2a^{-2}$
- B) $2a^2$
- C) $4a$
- D) $-4a^{\frac{2}{3}}$

Question 5.

[Tg:@abusat_760] If $8^x = 4$ and $3^{x+y} = 9$, what is the value of y ?

- A) -2

B) $\frac{2}{3}$

C) $\frac{4}{3}$

D) 2

Question 6.

[Tg:@abusat_760]

$$\frac{5(x+1)-6}{x} = 2$$

What value of x satisfies the equation above?**Question 7.**[Tg:@abusat_760] Which of the following is equivalent to the expression $-4x^2 + 12x$?

A) $-4(-x^2 + 3x)$

B) $-4x(x + 3)$

C) $-2x(2x - 6)$

D) $x(4x - 12)$

Question 8.

[Tg:@abusat_760] At a certain point in time, 1 US dollar was equivalent to 17.63 L.E. Based on this relationship, what was the value of 30 US dollars in L.E at that time?

A) 0.59

B) 17.63

C) 52.89

D) 528.9

Question 9.

[Tg:@abusat_760]

$$2(5 - 3x) = x + 31$$

Which of the following is the solution to the equation above?

A) -3

B) $-\frac{7}{3}$

C) $\frac{7}{3}$

D) 3

Questions 10-12 refer to the following information.

A fitness club is hosting an indoor triathlon. Each triathlon participant spends 5 minutes swimming in the pool, 40 minutes biking on a stationary bike, and 15 minutes running on a treadmill. Scores are given based on total combined distance in miles, D , of the three activities, which can be found using the equation below:

$$D = \frac{1}{12}s + \frac{2}{3}b + \frac{1}{4}r$$

In the equation, s is the participant's average swimming speed, in miles per hour (mph); b is the participant's average biking speed, in mph; and r is the participant's average running speed, in mph.

Question 10.

[Tg:@abusat_760] In the swim portion of the triathlon, Ahmed's average swimming speed was 1.44 miles per hour. His goal is to have a total combined distance of at least 11.2 miles. He knows he will have an average biking speed that is between 3 and 4 times faster than his average running speed. Which system of inequalities represents the ways in which Ahmed can meet his goal?

- A) $\frac{2}{3}b + \frac{1}{4}r \geq 11.08$
 $2b < r < 3b$
- B) $\frac{2}{3}b + \frac{1}{4}r \geq 11.08$
 $3r < b < 4r$
- C) $\frac{2}{3}b + \frac{1}{4}r \geq 11.2$
 $3b < r < 4b$
- D) $\frac{2}{3}b + \frac{1}{4}r \geq 11.2$
 $3r < b < 4r$

Question 11.

[Tg:@abusat_760] Amr's average running speed during the triathlon was 9.2 mph, and his average biking speed was 14 mph. His total combined distance was 11.75 miles. At what average speed, in mph, did he swim?

- A) 0.25
- B) 1.20
- C) 1.40
- D) 3.08

Question 12.

[Tg:@abusat_760] In the triathlon, Ghada ran exactly 8 times the distance she swam. Which equation relates her average swimming speed to her average running speed?

- A) $r = 2s$
- B) $r = 4s$

C) $r = \frac{8s}{3}$

D) $r = \frac{s}{8}$

Question 13.

[Tg:@abusat_760]

$$K(5 - 4y) = 7 - 8y$$

In the equation above, K is a constant. If no value of y satisfies the equation, what is the value of K ?

A) 3

B) 2

C) 1

D) 0

Question 14.[Tg:@abusat_760] Which of the following is equivalent to $\frac{x^2 + 3x + 2}{x + 3}$?

A) $3x$

B) $x + 2$

C) $x + \frac{2x}{x + 3}$

D) $x + \frac{2}{x + 3}$

Question 15.

[Tg:@abusat_760]

$$1.75c + 1.40n = t$$

The equation above gives the total cost t , in dollars, to purchase c pounds of cherries and n pounds of nectarines at a local grocery store. How much greater would be the total cost, in dollars, to purchase 4 pounds of cherries and 6 pounds of nectarines than to purchase 5 pounds of cherries and 2 pounds of nectarines? (Disregard the \$ sign when gridding your answer. For example, if your answer is \$8.47, grid 8.47)

Question 16.[Tg:@abusat_760] Which of the following is equivalent to the product of $-2M$ and $M - 1$?

A) $-2M^2 - M$

B) $-2M^2 + 2M$

C) $-2M^2 + 2$

D) -2

Question 17.**[Tg:@abusat_760]**

$$y = 3x + 5$$

Given the equation above, if $y = 15$, what is the value of x ?

Question 18.**[Tg:@abusat_760]** If $A + B = \sqrt{2(AB + 5)}$, what is the value of $A^2 + B^2$?**Question 19.****[Tg:@abusat_760]** If $5x + 8 = 8 - 5x$, what is the value of x ?

- A) 0
- B) 2
- C) 3
- D) 5

Question 20.**[Tg:@abusat_760]** If $N(x) = 10 + 2(x - 1) + x^2$, what is the value of $N(3)$?

- A) 16
- B) 23
- C) 37
- D) 43

Question 21.**[Tg:@abusat_760]** If the expression $3x + 12$ is equivalent to $3(x - a)$, where a is a constant, what is the value of a ?

- A) -3
- B) 3
- C) -4
- D) $\frac{10}{3}$

Question 22.**[Tg:@abusat_760]** If $\frac{40}{B - 5} = 5$, what is the value of $B - 5$?

- A) 3
- B) 4
- C) 5
- D) 8

Question 23.

[Tg:@abusat_760] What is the solution of $9K - 18 = -14$?

- A) -32
- B) $-\frac{32}{9}$
- C) $\frac{4}{9}$
- D) 4

Question 24.

[Tg:@abusat_760]

$$4(1 - x) = -4x + P$$

In the equation above, P is a constant. If the equation has no solution, which of the following must be true?

- A) $P \neq 4$
- B) $P \neq 1$
- C) $P \neq -1$
- D) $P \neq -3$

Question 25.

[Tg:@abusat_760] Which of the following is equivalent to $x^{\frac{2}{7}}$, where $x > 0$?

- A) $\frac{x^2}{x^7}$
- B) $x^2 - x^7$
- C) $\sqrt{x^7}$
- D) $(\sqrt[7]{x})^2$

Question 26.

[Tg:@abusat_760] Which of the following is equivalent to $(3x - \frac{1}{2})(x - \frac{1}{3})$?

- A) $3x^2 - \frac{3}{2}x - \frac{1}{6}$
- B) $3x^2 - \frac{3}{2}x + \frac{1}{6}$
- C) $3x^2 - \frac{9}{2}x - \frac{1}{6}$
- D) $3x^2 - \frac{11}{6}x + \frac{1}{6}$

Question 27.**[Tg:@abusat_760]**

$$M(x^{13} - x^{11}) = x^{12} + x^{11}$$

Based on the condition $x > 1$, what is M in terms of x ?

- A) $x - 1$
- B) $x + 1$
- C) $\frac{1}{x - 1}$
- D) $\frac{1}{x + 1}$

Question 28.**[Tg:@abusat_760]**

$$p = (3v - 6)$$

$$q = (3v + 6)$$

What is $\frac{pq}{9}$ in terms of v ?

- A) $v^2 - 4$
- B) $v^2 - 36$
- C) $9v^2 - 4$
- D) $9v^2 - 36$

Question 29.

[Tg:@abusat_760] A recipe that serves 5 people calls for b bananas and d dates. According to this recipe, what is the total number of bananas and dates needed to serve 25 people?

- A) $b + d$
- B) $5b + 5d$
- C) $20b + 20d$
- D) $25b + 25d$

Question 30.**[Tg:@abusat_760]**

$$\frac{3}{x^2 - 4} = \frac{1}{x - 2} - \frac{1}{x + 2}$$

Which statement describes the solution to the equation above?

- A) 2 is the only solution.
- B) -2 is the only solution.
- C) 2 and -2 are both solutions.

D) There are no solutions.

Question 31.

[Tg:@abusat_760]

$$\frac{1}{3}(3ax - 6) = 4$$

Based on the equation above, what is the value of $5ax$?

Questions 32-36 refer to the following information.

The following table shows several values of function F and function G :

x	$F(x)$	$G(x)$
1	5	8
3	8	10
6	10	0
10	6	6
15	3	-2

Question 32.

[Tg:@abusat_760] According to the table, if $F(6) = a$, what is $G(a)$?

Question 33.

[Tg:@abusat_760] According to the table, if $G(3) = b$, what is $F(b)$?

Question 34.

[Tg:@abusat_760] According to the table, what is $G(F(6))$?

Question 35.

[Tg:@abusat_760] According to the table, what is $(G + F)(15)$?

Question 36.

[Tg:@abusat_760] According to the table, what is $(G - F)(1)$?

Question 37.

[Tg:@abusat_760]

$$x = 3a + b$$

$$y = a - b$$

Given the two equations above, which choice is equivalent to xy ?

- A) $3a^2 - b^2$
- B) $3a^2 - 4ab - b^2$
- C) $3a^2 - 2ab - b^2$
- D) $3a^2 - 2ab + b^2$

Question 38.

[Tg:@abusat_760] Which of the following is equivalent to $\frac{x^3}{x} + \frac{x^5}{x^3}$ for all positive values of x ?

- A) $2x^2$
- B) $2x^4$
- C) $2x^6$
- D) $2x^8$

Question 39.

[Tg:@abusat_760] Which of the following expressions is equivalent to $\frac{a}{b}(a-b)(a+b)$?

- A) $\frac{a^3}{b} - ab$
- B) $\frac{a^2}{b} - b$
- C) $\frac{a^3}{b} + \frac{2a^2}{b} + ab$
- D) $\frac{a^2}{b} - ab$

Question 40.

[Tg:@abusat_760] If $u = x + 2$ and $u \geq 0$, which of the following is equivalent to $(x-1)^2\sqrt{x+2}$?

- A) $(u-3)^2\sqrt{u}$
- B) $(u-2)^2\sqrt{u}$
- C) $(u-1)^2\sqrt{u}$
- D) $(u+1)^2\sqrt{u}$

Question 41.

[Tg:@abusat_760]

$$y = 2x^2 + 2x + 1$$

$$y = b$$

When the equations above are graphed in the xy -plane, they form a parabola and a line. If the line intersects the parabola in exactly one point, what is the value of b ?

- A) $-\frac{1}{2}$
- B) $-\frac{1}{4}$
- C) $\frac{1}{2}$

D) 1

Question 42.

[Tg:@abusat_760] Which of the following expressions is equivalent to $\frac{x-3}{x^2-1}$?

A) $\frac{2}{x+1} - \frac{1}{x-1}$

B) $\frac{1}{x+1} - \frac{2}{x-1}$

C) $\frac{2}{x+1} + \frac{1}{x-1}$

D) $\frac{1}{x+1} + \frac{2}{x-1}$

Question 43.

[Tg:@abusat_760] If $a = 2^{3k}$ and $b = 4^{-3k}$, which of the following is equivalent to $\frac{b}{a}$?

A) 2^{-9k}

B) 2^{-6k}

C) 2^{-3k}

D) 2^0

Question 44.

[Tg:@abusat_760]

$$\frac{(x+1)(x-2)}{2} + 3x$$

If the expression above is rewritten in the form $ax^2 + bx + c$, where a , b , and c are constants, what is the value of b ?

A) 2.5

B) 3.5

C) 4

D) 5

Question 45.

[Tg:@abusat_760] Which choice is equivalent to $\frac{x^{-\frac{1}{3}}y^3}{y^{-2}}$?

A) $\frac{y}{\sqrt[3]{x}}$

B) $\frac{y}{x^3}$

C) $\frac{y^5}{\sqrt[3]{x}}$

D) $\frac{y^5}{x^3}$

Question 46.

[Tg:@abusat_760] If $a = b + 4$, what is the value of $a^2 - 2ab + b^2$?

Question 47.

[Tg:@abusat_760] Which of the following is the simplified form of $(3x^2 - 4x + 2) - (4x^2 - 3x - 1)$?

- A) $-x^2 - x + 3$
- B) $-x^2 - 7x + 1$
- C) $x^2 - 7x + 3$
- D) $-x^2 - x + 1$

Question 48.

[Tg:@abusat_760] If $a = \frac{1}{b}$ and $a + b = 0$, what is the value of a ?

- A) $\sqrt{2}$
- B) $2\sqrt{2}$
- C) 4
- D) The value of a cannot be a real number.

Question 49.

[Tg:@abusat_760] Which of the following functions corresponds to a graph in the xy -plane with x -intercepts at -2 and 3, and a y -intercept of 12?

- A) $f(x) = (-2x - 4)(2x - 6)$
- B) $f(x) = (-2x - 4)(x - 3)$
- C) $f(x) = (4x + 2)(3x - 9)$
- D) $f(x) = (2x + 4)(2x + 6)$

Question 50.

[Tg:@abusat_760] Which choice represents the values of x for which the function $y = \frac{x^2 + 2x - 3}{x^2 + 5x - 6}$ is undefined?

- A) $x = -1, x = 6$
- B) $x = -2, x = 3$
- C) $x = 1, x = -6$
- D) $x = 2, x = -3$

Answer Sheet

1	2	3	4	5	6	7	8	9	10
C	B	D	A	C	1/3	C	D	A	B

11	12	13	14	15	16	17	18	19	20
C	C	B	D	3.85	B	10/3	10	A	B

21	22	23	24	25	26	27	28	29	30
C	D	C	A	D	B	C	A	B	D

31	32	33	34	35	36	37	38	39	40
30	6	6	6	1	3	C	A	A	A

41	42	43	44	45	46	47	48	49	50
C	A	A	5/2	C	16	A	D	B	C